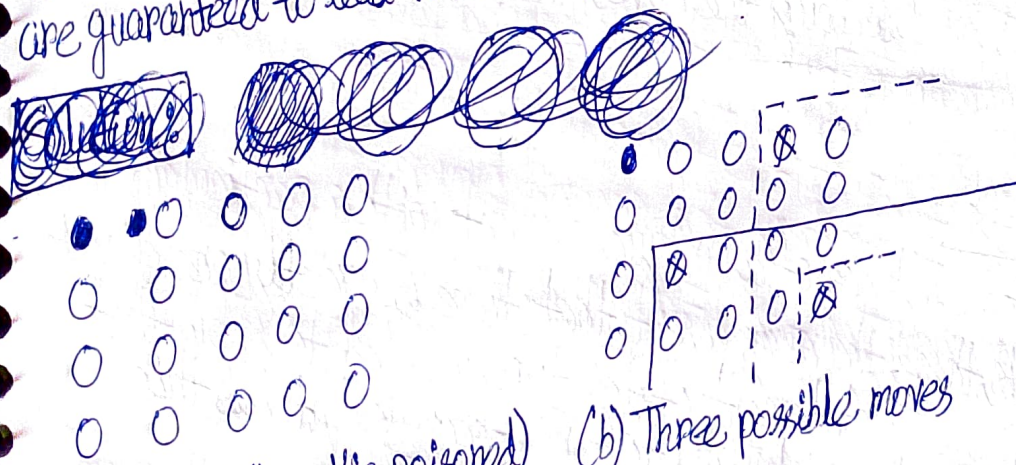


which of these two pairs works! $\square$

Example 12 illustrates the subtlety of nonconstructive existence proofs

Example 12: **Chomp** is a game played by two players. In this game, cookies are laid out on a rectangular grid. The cookie in the top left is poisoned, as shown in Figure 1(a). The two players take turns making moves; at each move, a player is required to eat a remaining cookie, together with all cookies to the right and/or below it (see Figure 1(b), for example). The loser is the player who has no choice but to eat the poisoned cookie. We ask whether one of the two players has a winning strategy. That is, can one of the players always make moves that are guaranteed to lead to a win?



(a) Chomp (top left cookie poisoned) (b) Three possible moves

Solution: We will give a nonconstructive existence proof of a winning strategy for the first player. That is, we will show that the first player always has a winning strategy without explicitly describing the moves this player must follow.

First, note that the game ends and cannot finish in a draw because with each move at least one cookie is eaten, so after no more than $m \times n$ moves, the game ends, where the initial grid is $m \times n$.

Now suppose that the first player begins the game by eating just the cookie in the bottom right corner. There are two possibilities, this is the first move of a winning strategy of the first player, or the second player can make a move that is the first move of the winning